# F79SP Stochastic Processes 1 2024-25
# **Solutions** to Exam Qns, December 2024.

1. **Question 1 corresponds to CS2 Syllabus (2022) Objectives 3.1.1 and 3.1.2.**

**Questions of this type were discussed in lectures and tutorials.**

(a), This is a coursework.

We may represent $T_a \equiv T_{a,0}$ as a sum of independent random variables

$$T_{a,0} = T_{a,a-1} + T_{a-1,a-2} + \ldots + T_{1,0}$$

where $T_{a,a-1}$ is a (random) time to get $a-1$ starting from $a$, $T_{a-1,a-2}$ is a consecutive time to get $a-2$ from $a-1$ etc. Due to the space translation invariance (discussed in detail in lectures), these random variables are identically distributed. Therefore

$$\mathbf{E}T_{a,0} = a\mathbf{E}T_{1,0} \equiv a\mathbf{E}T_1.$$

**(2 marks)**

Further, use the total probability formula in one step:

$$T_1 = 1 + \mathbf{I}(X_1 = 2)\widetilde{T}_2$$

where the indicator function $\mathbf{I}(A)$ equals 1 if event $A$ occurs and 0, otherwise, and $\widetilde{T}_2$ is the time to get 0 for the simple random walk that stars from the state 2 at time 1. Due to independence,

$$\mathbf{E}T_1 = 1 + p\mathbf{E}\widetilde{T}_2 = 1 + p\mathbf{E}T_2 = 1 + 2p\mathbf{E}T_1$$

where the second equality follows from the time homogeneity.

**(2 marks)**

Solving the equation gives $\mathbf{E}T_1 = 1/(1-2p)$, and then the result for any $a$ follows.

**(2 marks)**

(b) Here we introduce similar notation for $T_{j,j-1}$ and get again the formula

$$\mathbf{E}T_a = a\mathbf{E}T_1.$$

But then the total probability formula for $T_1$ gives us:

$$T_1 = 1 + \mathbf{I}(X_1 = 2)\widetilde{T}_2 + \mathbf{I}(X_1 = 1)\widetilde{T}_1$$

where $\widetilde{T}_1$ is the time to get 0 for the simple random walk that starts from 1 at time 1, and $\widetilde{T}_2$ is as before. Then

$$\mathbf{E}T_1 = 1 + 2p\mathbf{E}T_1 + r\mathbf{E}T_1.$$

**(2marks)**

Multiplying the LHS by $p + r + q = 1$ and opening then the parentheses, we get

$$(q-p)\mathbf{E}T_1 = 1, \quad \text{and} \quad \mathbf{E}T_1 = 1/(q-p).$$

Then the general result follows.

**(2 marks)**

(c)(i) Here $a = 5$ and $\xi_i = 1 - Y_i$, so $p = 1/20$, $r = 7/20$, and $q = 12/20$. So we may apply the result from (b) to get

$$\mathbf{E}T_5 = \frac{5}{11/20} = 100/11 \approx 9,09.$$

**(3 marks)**

(c)(ii) $Y_i \geq \widetilde{Y}_i = \min(Y_i, 2)$ which correspond to the simple random walk $\widetilde{S}_n = 5 + \sum_1^n \widetilde{\xi}_i$ where $\widetilde{\xi}_i = 1 - \widetilde{Y}_i$. Then $\widetilde{S}_n \geq S_n$, and the ruin time for $S_n$ is smaller than that for $\widetilde{S}_n$.

**(2 marks)**

Then

$$\mathbf{E}T_5 \leq \frac{5}{9/20} = 100/9 \approx 11,1.$$

**(1 mark)**

2. **Question 2 corresponds to CS2 Syllabus (2022) Objectives 3.2.2 and 3.2.3.**

Questions of this kind were discussed in lectures and tutorials.

(a). This is again a coursework. We have to prove three statements: (a1) that this is a limiting distribution, (a2) that this a stationary distribution, and (a3) that this is the only stationary distribution.

(a1) We know that, for any $i$,

$$\sum_j p_{i,j}^{(n)} = 1.$$

Letting $n$ to infinity and noting that there is a finite number of summands, we can interchange the sum and the limits to get

$$\sum_j \omega_j = 1.$$

(see the Hint to the question for definition of the $\omega$'s in the Exam paper).
**(2 marks)**

(a2). For any $i$ and $j$, in the Chapman-Kolmogorov equation,

$$p_{i,j}^{(n+1)} = \sum_k p_{i,k}^{(n)} p_{k,j},$$

we let $n$ tend to infinity. Since there is a finite number of summands, the limit of the sum is the sum of the limits, and we get

$$\omega_j = \sum_k \omega_k p_{k,j}.$$

So this is a stationary distribution.
**(2 marks)**

(a3). Let $\pi = \{\pi_j\}$ be any stationary distribution. Then, from the property of a stationary distribution, for all $n$,

$$\pi_j = \sum_i \pi_i p_{i,j}^{(n)}.$$

Letting $n$ to infinity and noticing again that the sum is finite, we get

$$\pi_j = \sum_i \pi_i \omega_j = \omega_j \sum_i \pi_i = \omega_j$$

since $\sum_i \pi_i = 1$.
**(3 marks)**

(b)(i) The transition matrix is

$$P = \begin{pmatrix} 0.11 & 0.89 & 0 & 0 \\ 0.11 & 0 & 0.89 & 0 \\ 0.01 & 0.1 & 0 & 0.89 \\ 0 & 0.01 & 0.1 & 0.89 \end{pmatrix}$$

**(2 marks)**

(b)(ii) The Markov chain has 4 states, they are 0,1,2, and 3.

First, any state communicates with each other: one can get any state from 0, and one can get 0 from any state: $p_{0,1} > 0, p^{(2)}_{0,2} = 0.89^2 > 0, p^{(3)}_{0,3} = 0.89^3 > 0$ and $p_{1,0} > 0, p_{2,0} > 0, p^2_{3,0} > p_{3,1}p_{1,0} > 0$.

**(1 mark)**

Second, one can take matrix $P^3$ (or matrix $P^4$) and find that all its entries are strictly positive. This leads to aperiodicity. Another, even simpler proof of aperiodicity follows from observing that there are non-zero entries on the main diagonal (for example, $p_{0,0} = 0.11 > 0$). Note that students are supposed to produce only one of the arguments above.

**(1 mark)**

(b)(iii) Due to (a), there is only one stationary distribution which coincides with any limiting distribution. We get 4 equations:

$$\begin{aligned} \pi_0 \cdot 0.11 &+ \pi_1 \cdot 0.11 + \pi_2 \cdot 0.01 + 0 = \pi_0 \\ \pi_0 \cdot 0.89 &+ 0 + \pi_2 \cdot 0.1 + \pi_3 \cdot 0.01 = \pi_1 \\ 0 &+ \pi_1 \cdot 0.89 + 0 + \pi_3 \cdot 0.1 = \pi_2 \\ 0 &+ 0 + \pi_2 \cdot 0.89 + \pi_3 \cdot 0.89 = \pi_3 \end{aligned}$$

together with the normalising equality $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$.

**(3 marks)**

This system of equations has solution

$$\pi_0 \approx 0.004, \pi_1 \approx 0.023, \pi_2 \approx 0.107, \pi_3 \approx 0.869.$$

**(1 mark)**

(b)(iv)(A) The new transition matrix is

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.11 & 0 & 0.89 & 0 \\ 0.01 & 0.1 & 0 & 0.89 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

**(2 marks)**

So we have two absorbing states/classes, 0 and 3, and one open (non-closed) class $\{1, 2\}$. So all stationary distributions have the form

$$(\gamma_1, 0, 0, \gamma_2)$$

where $\gamma_1 + \gamma_2 = 1$ and $\gamma_1, \gamma_2 \geq 0$.
**(2 marks)**

(b)(iv)(B) Let $C_1 = \{0\}$ and $C_2 = \{3\}$ and let $\alpha_i = \alpha_i^{(C_1)}$ be the absorption probabilities in class $C_1$. Clearly, $\alpha_0 = 1$ and $\alpha_3 = 0$. Further, $\alpha_1 = 0.11 + 0.89\alpha_2$ and $\alpha_2 = 0.01 + 0.1\alpha_1$. Then $0.911\alpha_2 = 0.21$, and $\alpha_2 = 21/911$ and $\alpha_1 = 1189/9110$.
**(2 marks)**

Further, $\alpha_1^{(C_2)} = 1 - \alpha_1 = 7921/9110$ and $\alpha_2^{(C_2)} = 1 - \alpha_2 = 890/911$.
**(2 marks)**

3. **Question 3 corresponds to CS2 Syllabus (2022) Objectives 3.2.2 and 3.2.3.**

Questions of this kind were discussed in lectures and tutorials.

(a). Let $\mu$ be the mean number of offspring of a single individual. Inequality $\mu \leq 1$ is equivalent to $\varepsilon(1, p) = 1$ which is equivalent to $\varepsilon(k, p) = (\varepsilon(1, p))^k = 1$.
**(1 mark)**

This occurs iff $(1 + 2p)/4 + 2 \cdot p/4 \leq 1$ and iff $3(1 - p) \geq p$, i.e. $p \leq 3/4$.
**(2 marks)**

(b) This is when $\varepsilon(k, p) = (\varepsilon(1, p))^k = 0$. For this, we need $p_0 = 0$, i.e. $p = 1$.
**(2 marks)**

(c) There are two options, either
– solving the quadratic equation $p_0 + p_1\varepsilon + p_2\varepsilon^2 = \varepsilon$ and taking the smaller root, or
– using the fact that $\varepsilon$ equals the ruin probability.

The latter is simpler. If $p = 5/6$, then $p_0 = 1/8$ and $p_2 = 5/24$,
**(2 marks)**

so the extinction probability is $(p_0/p_2)^k$ which is $3/5$ if $k = 1$
**(2 marks)**

and $(3/5)^5$ if $k = 5$.
**(2 marks)**

(d) If $p = 0$ then $p_2 = 0$, $p_1 = 1/4$ and $p_0 = 3/4$. Then
$T_1$ has a geometric distribution with parameter $3/4$,
**(1 mark)**

so $\mathbf{E}T_1 = 4/3$ and $\mathbf{P}(T_1 > 3) = (1/4)^3 = 1/64$.

**(2 marks)**

4. **Question 4 corresponds to CS2 Syllabus (2022) Objectives 3.3.1 and 3.3.2.**

Questions of this type were discussed in lectures and tutorials.

(a) $X \sim Pois(\lambda \cdot \frac{1}{2} \cdot 4) = Pois(10)$.

**(1 mark)**

(b) Let $Y$ be the number of claims of size 20 during the first day, and let $Z$ be the total number of claims within four days. We know that $Y$ has a Poisson distribution with parameter 5/4 while $Z$ has a Poisson distribution with parameter 20.

**(2 marks)**

Then the conditional distribution of $Y$ given $Z = 5$ is Binomial with parameters $n = 5$ and $p = \frac{5/4}{20} = \frac{1}{16}$, so

$$\mathbf{P}(Y = 3 \mid Z = 5) = \frac{5!}{3!2!} \cdot p^3(1-p)^2 \approx 0.002146.$$

**(3 marks)**

(c) We count here only claims of sizes 20 and 120. Since claims of size 120 occur with double intensity in comparison with claims of size 20, each claim (either 20 or 120) is of size 120 with probability 2/3. The probability of interest is the probability to have at least 5 "successes" in 5+2-1=6 experiences,

**(2 marks)**

which is

$$6 \cdot (2/3)^5 \cdot (1/3) + (2/3)^6 = \frac{256}{6561} \approx 0.35.$$

**(2 marks)**

(d) This is the sum of three probabilities: **P** (no claims at all) plus **P** (between 1 and 3 claims of size 20 and no other claims) plus **P** (1 claim of size 70 and no other claims),

**(2 marks)**

which is

$$e^{-10} + \left(\frac{5}{2} + \frac{(5/2)^2}{2!} + \frac{(5/2)^3}{3!}\right) e^{-10} + \frac{5}{2}e^{-10} = \frac{515}{48} \cdot e^{-10} \approx 10.73 \cdot e^{-10}.$$

**(3 marks)**

5. **Question 5 corresponds to CS2 Syllabus (2022) Objectives 3.3.1, 3.3.3 and 3.3.4.**
Questions of this type were discussed in lectures and tutorials.

(a)

(i) There are three states, all states intercommunicate since all off-diagonal elements of the matrix are positive. The total number of states is finite. Therefore there is only one limiting distribution (for all initial states) that coincides with the unique stationary distribution.

**(2 marks)**

To find it, we have to solve the system of equations:

$$\pi Q = \mathbf{0}, \quad \sum_i \pi_i = 1$$

that leads to the solution

$$\pi_1 = 1/5, \pi_2 = 7/15, \pi_3 = 1/3.$$

**(2 marks)**

(ii) The transition matrix for the embedded Markov chain is

$$P^* = \begin{pmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

Again, all states inter communicate. In addition, this Markov chain is **aperiodic**: $p^2_{22} = p_{21}p_{12} > 0$ and $p^3_{22} > p_{23}p_{31}p_{12} > 0$. Therefore there is only one limiting distribution (for all initial states) that coincides with the unique stationary distribution.

**(2 marks)**

We have to solve the system of equations:

$$\pi^* P^* = \pi^*, \sum \pi^*_i = 1$$

that leads to the solution

$$\pi^*_1 = 1/3, \pi^*_2 = 7/18, \pi^*_3 = 5/18.$$

**(2 marks)**

(iii) All answers do not depend on $\lambda$. Clearly, $\lambda$ is a scaling parameter (for example, we may count time in minutes or in hours or in other units),

and we know that the stationary probability at each state is a fraction of time the Markov chain spends there. Clearly, the fraction does not depend on the scaling.

**(1 mark)**

(b) This is again a transition-rate matrix with all non-zero non-diagonal elements. Due to the same reasons, both continuous-time Markov process and discrete-time Markov chain have unique stationary = limiting distributions.

**(1 mark)**

Now we may observe that this matrix is shift-invariant: shifting $1 \to 2, 2 \to 3, 3 \to 1$ keeps the matrix the same. Therefore all states have equal "chances", and all discrete- and continuous-time stationary probabilities are $1/3$.

**(2 marks)**